

Shafin Mahamud

Bangladeshi | [Gmail](#) | (+880)-1533-764560 | [LinkedIn](#) | GitHub: [shafin027](#)

EDUCATION

Brac University

B.S. in Computer Science

Merul Badda, Dhaka -1212, Bangladesh.

Expected Graduation, May 2027

- o **Concentrations:** Data Science, Machine Learning, Artificial Intelligence
- o **CGPA:** 3.61/4.00.
- o **Related Coursework:** Neural Network Processing, Computer Networks, Computer Operating System, Artificial Intelligence, Data Structures, Algorithms, Discrete Mathematics, Object-Oriented Programming, Data Communications, Statistics and Probability, Advanced Probability & Stochastic Process Analysis.
- o **Activities:** Runner-up (RS-64 Cricket Tournament), Champion (RS Cup Cricket 2025), Main Team Player (Brac University Cricket Team).

Dhaka Residential Model College

Higher Secondary Certificate (HSC) – Science Department

Mohammadpur, Dhaka -1207, Bangladesh.

July 2018- April 2020

- o **Concentrations:** Physics, Chemistry, Mathematics, Biology, and ICT.
- o **GPA:** 5.00/5.00.
- o **Activities:** Organized DRMC Science Fest, Core Member, DRMC IT Club (Organizer of the first DRMC International IT Carnival), Played for House Cricket Team, Active participant in campus sports events across the 52-acre campus.

EXPERIENCE

BayShore Communication

Cloud Engineering Intern

Aftab Nagar, Dhaka -1212.

Sept. 2024 – Present

- Assisted in deploying and managing AWS services (EC2, S3, CloudWatch) for cloud infrastructure.
- Monitored system performance and resolved basic cloud environment issues.
- Supported troubleshooting and resolving basic cloud environment issues under the guidance of senior engineers.
- Documented deployment processes and gained hands-on experience with version control tools like Git.

PROJECTS

[Annual Income Prediction](#)

Project Developer

Merul Badda, Dhaka -1212.

Aug. 2025 – Sept. 2025

- Classification problem to predict if an individual's annual income is $> \$50K$ or $\leq \$50K$ using the Adult Income dataset.
- The project trained and tested three supervised machine learning models (Naive Bayes, Logistic Regression, and Neural Network) and used K-Means Clustering for an unsupervised analysis.

[Cattle Skin Disease Prediction](#)

Project Developer

Merul Badda, Dhaka -1212.

Jan. 2026 – Apr. 2026

- Developed a high-accuracy cattle skin disease classification system using computer vision and deep learning.
- Implemented a self-supervised SimCLR fine-tuned model, achieving 95.48% test accuracy for disease prediction from cattle skin images.
- Compared CNN-based architectures including DenseNet121, ResNet50, EfficientNetB0, and SimCLR using Accuracy, F1-score, ROC-AUC, and Confusion Matrix analysis.

[Multi-Zone Enterprise Network Design](#)

Project Developer

Merul Badda, Dhaka -1212.

Apr. 2026 – May. 2026

- Designed a six-zone enterprise network in Cisco Packet Tracer using VLSM subnetting from 10.109.0.0/16.
- Configured hybrid routing with RIP v2, static routes, default routes, and floating static routes.
- Implemented router-based and server-based DHCP, centralized DNS, HTTP web service, and SMTP/POP3 email.
- Verified full connectivity using ping, tracert, show ip route, DNS lookup, web access, and email testing.

Multi-Class News Headline Classification

Merul Badda, Dhaka -1212.

Project Developer

Apr. 2026 – May. 2026

- Developed an end-to-end NLP pipeline to classify news headlines into Business, Science & Technology, Sports, and World News categories.
- Implemented TF-IDF and Skip-gram (Word2Vec) representations with Logistic Regression, DNN, RNN, GRU, LSTM, and Bidirectional LSTM models.
- Performed extensive EDA, preprocessing comparison, hyperparameter tuning, and evaluation using Accuracy, Macro F1-score, ROC-AUC, and Confusion Matrix analysis.

GreenLoop - Sustainable Waste Management Ecosystem

Merul Badda, Dhaka -1212.

Project Developer

Apr. 2026 – May. 2026

- Developed a full-stack MERN-based sustainability platform connecting households, waste collectors, recycling centers, and businesses in a unified recycling ecosystem.
- Implemented real-time location tracking, waste pickup scheduling, eco-points rewards, community engagement, and carbon credit marketplace features.
- Built role-based dashboards, proof-image verification, fraud detection logic, and sustainability analytics using React, Node.js, Express.js, and MongoDB.

xv6-riscv-expert — Claude Code Skill

Merul Badda, Dhaka -1212.

Developer

Apr. 2026 – May. 2026

- Developed an AI-assisted Claude Code skill for solving and debugging xv6-riscv operating system assignments, focusing on syscalls, schedulers, virtual memory, traps, and kernel debugging.
- Built structured workflows for RISC-V kernel development, trap analysis, page-table debugging, and scheduler implementation using xv6 architecture concepts.
- Published and maintained an open-source developer tool with GitHub documentation, automated installation support, and debugging guidance for OS students and systems programmers.

ACTIVITIES AND LEADERSHIP

Aspire Institute Inc. (spin-off from Harvard University)

New York, USA.

ALP Community Moderator

Dec 2024 – Present

- Managed and engaged a community of 200+ selected participants from the Aspire Leadership Program.
- Communicated key announcements, program updates, and deadlines to ensure smooth participation.
- Responded to queries and provided support, leveraging alumni experience to guide participants.
- Fostered a collaborative environment by encouraging discussions and peer engagement.
- Coordinated with program administrators to address participant concerns and improve community management.

SKILLS

Programming: Python, Java, HTML/CSS, PHP, C.

Tools: Visual Studio Code, PyCharm, AWS, Jupyter Notebooks, Kaggle, and Git.